

LEARNING THEORIES



Notes

- 7.1 *Introduction*
- 7.2 *Learning Outcomes*
- 7.3 *Brief Ideas on Learning Theories & their Implications in Teaching Learning Process*
- 7.4 *Learning By Doing – The Power of Experiential Learning*
- 7.5 *Playful Classrooms – Toy-Based Pedagogy For Joyful Learning*
- 7.6 *Foundations of Integrated Learning*
- 7.7 *Fostering Learner Autonomy in Primary Classrooms*
- 7.8 *Facilitating Child Centred and Flexible Pedagogy-Teacher as Facilitator And Co-Explorer*
- 7.9 *Let Us Sum Up*
- 7.10 *End Exercises*
- 7.11 *References*

7.1 INTRODUCTION

The early years of education form the foundation for lifelong learning, well-being, and holistic development. In keeping with the vision of the National Education Policy 2020 (NEP 2020) and the National Curriculum Framework for the Foundational Stage (NCF-FS, 2022), there is a growing recognition that classroom practices must shift from rote, textbook-driven instruction to pedagogies that are experiential, engaging, inclusive, and joyful. This chapter has been developed to equip primary teachers with the knowledge and tools to translate these progressive curricular aims into classroom realities. It introduces

Process of Learning**Notes**

teachers to a range of effective and developmentally appropriate learning approaches, including experiential learning, toy-based pedagogy, art-integrated learning, guided inquiry, problem-solving, promoting learner autonomy and redefining the teacher's role as a facilitator and co-explorer. Each perspective is aligned with India's policy priorities and backed by contemporary research in child development and pedagogy. The chapter includes hands-on activities, sample lesson ideas, and age-appropriate strategies to make these approaches practical and replicable in diverse classroom settings. Moreover, the chapter encourages teachers to reflect on their own roles and adapt to child-centric, flexible, and inclusive pedagogies that recognize each learner's context and potential. Teachers will also learn to design learning experiences that are culturally rooted, locally relevant, and suited to foundational learning goals outlined in the NIPUN Bharat Mission. This chapter is intended for pedagogical transformation—empowering teachers to build classrooms where children actively construct knowledge, engage in meaningful learning experiences, and develop into confident, creative, and compassionate individuals.

**7.2 LEARNING OUTCOMES**

After completion of this chapter, the learner:

- designs experiential learning activities aligned with curriculum and applies them effectively in classroom contexts;
- integrates active pedagogies like toy-based learning and art integrated learning to boost engagement, retention, and holistic development;
- applies guided inquiry and problem-solving strategies through well-crafted questions to enhance conceptual understanding;
- promotes learner autonomy by using strategies that support independent learning, goal setting, and self-regulation;
- facilitates a learner-centered environment by shifting from direct instruction to practices that encourage inquiry, dialogue, and critical thinking.

7.3 BRIEF IDEAS ON LEARNING THEORIES & THEIR IMPLICATIONS IN TEACHING LEARNING PROCESS

Learning theories are foundational tools that help educators understand how students learn and develop. These theories inform classroom practices by explaining the cognitive, emotional, and social aspects of learning, allowing teachers to tailor their instruction to better support student engagement and success (Ormrod, 2017; Schunk, 2020).

Process of Learning



Table 7.1: Categories of Learning theories and their key Characteristics

S. No.	Learning Theory	Key Theorists	Focus Area	Core Principles	Classroom Applications
1	Behaviourist	B.F. Skinner, Ivan Pavlov	Observable behaviour and external stimuli	Learning is shaped by reinforcement and punishment (conditioning)	Use of rewards, tokens, praise; classroom rules; drills; behaviour contracts.
2	Cognitive	Jean Piaget, Jerome Bruner	Mental processes (thinking, memory, reasoning)	Learning is an internal process of acquiring and organizing knowledge.	Concept mapping, scaffolding, developmental stage-based tasks, discovery learning.
3	Constructivist	Lev Vygotsky, Jean Piaget	Active knowledge construction through experience	Learners build understanding through reflection and social interaction.	Inquiry-based learning, group projects, hands-on activities, use of ZPD and scaffolding.
4	Social Learning	Albert Bandura	Learning through social context and observation	Behaviour is learned through observing, modelling, and imitation.	Peer tutoring, cooperative learning, role modelling, promoting self-efficacy.
5	Humanistic	Abraham Maslow, Carl Rogers	Holistic development and personal growth	Learning occurs when learners feel safe, valued, and motivated to self-actualize.	Emotionally supportive environment, student choice, creativity, open discussions.

Notes

Process of Learning

Notes

When applied together, these theories offer a comprehensive strategy for effective Preparatory & Foundational at teaching level. Behaviourism supports discipline and routine; cognitive and constructivist approaches enhance understanding and critical thinking; social learning builds cooperation; and humanistic theories nurture the emotional and motivational needs of learners (Ormrod, 2017; Woolfolk, 2019). A well-rounded Preparatory & Foundational classroom integrates all these elements, supporting the holistic development of each child.

**CHECK YOUR PROGRESS 7.1**

1. Which learning theory emphasizes the role of external reinforcement and punishment in shaping student behaviour?
 - a) Cognitive Learning Theory
 - b) Behaviourist Learning Theory
 - c) Constructivist Learning Theory
 - d) Humanistic Learning Theory
2. According to Vygotsky's constructivist theory, what is the Zone of Proximal Development (ZPD)?
 - a) The level at which a learner can solve problems independently without help
 - b) The range of tasks a learner can perform with guidance or collaboration
 - c) The final stage of cognitive development where abstract thinking occurs
 - d) A learning style focused on observing and imitating others

Ans. 1-b; 2-b

7.4 LEARNING BY DOING – THE POWER OF EXPERIENTIAL LEARNING

7.4.1 Making Learning Real: Principles of Experiential Learning

Experiential learning emphasizes active engagement through direct, hands-on experiences rather than passive reception of information. Learners construct

understanding by doing, reflecting, and applying knowledge in real-life contexts, which enhances both relevance and retention (Kolb, 1984). This approach bridges abstract concepts with tangible experiences, making learning more meaningful, especially for young children (Dewey, 1938; Kolb, 1984). Central to this method are hands-on activities that encourage exploration, manipulation of materials, and observation, allowing learners to form conclusions through experience (Beard & Wilson, 2013). In Preparatory education, such activities are particularly effective as they stimulate multiple senses, foster inquiry, and ground abstract ideas in everyday contexts (Kolb, 1984; Woolfolk, 2019).

Process of Learning



Notes

Table 7.2: Key Learning Theories Supporting Experiential and Hands-On Learning

Learning Theory	Proponent	Application in Concept Building
Experiential Learning Theory	David Kolb	Emphasizes the learning cycle: concrete experience → reflective observation → abstract conceptualization → active experimentation.
Constructivist Theory	Jean Piaget	Learners construct knowledge through active engagement and interaction with the environment.
Sociocultural Theory	Lev Vygotsky	Learning is mediated by tools, language, and social interaction; includes the Zone of Proximal Development (ZPD).
Multiple Intelligences Theory	Howard Gardner	Supports varied learning modalities like visual, bodily-kinesthetic, and logical-mathematical intelligence in activity design.
Bruner's Theory of Instruction	Jerome Bruner	Conceptual understanding builds through enactive (doing), iconic (seeing), and symbolic (thinking) stages.

Principles for Designing Effective Hands-On Activities

1. **Real-World Relevance:** Activities should connect with children's lives and environments.
2. **Active Engagement:** Learners should physically and mentally engage with the materials and concepts.

Process of Learning



Notes

3. Exploration and Inquiry: Encourage open-ended questioning and discovery.
4. Reflection: Include time for discussing what was learned and how.
5. Child-Centric: Allow flexibility, choices, and child-led variations.
6. Developmentally Appropriate: Activities should match the learners' age and cognitive level.

Table 7.3: Hands-On Activities Aligned with Learning Concepts & Theories

Subject	Activity	Steps	Learning theories aligned	
Science	States of Matter- "Melting Ice Explorers"	Provide ice cubes and various materials (metal spoon, plastic cup, cloth, salt). Ask: "Which item melts ice fastest?" Learners observe, record, and share findings.	Kolb's Experiential Learning	Concrete experience (touching and observing ice), Reflective observation (noticing melting speed), Conceptualization (understanding heat transfer), Experimentation (trying different items).
			Piaget's Constructivism	Learners discover scientific ideas through trial and error.
Mathematics	Multiplication as Repeated Addition- "Seed Arrays"	Learners create arrays using seeds (e.g., 3 rows of 4 seeds). They count and write multiplication sentences.	Gardner's Bodily-Kinesthetic and Logical-Mathematical Intelligences	Combining movement with abstract reasoning
			Vygotsky's Sociocultural Theory	Peer discussions support deeper understanding in the ZPD

Environmental Studies	Plant Life Cycle-“Grow a Bean Plant”	Each student plants a seed, observes growth over two weeks, and records changes using drawings or journals.	Kolb’s Cycle	Learners engage in repeated reflection and observation.
			Constructivism	Personal experience leads to understanding life processes.
Language Development	Story Sequencing-“Story Stones”	Provide painted stones with images (sun, child, house, tree). Children create and tell their own stories using the stones in sequence.	Gardner’s Linguistic and Visual-Spatial Intelligences	Combining visual aids with storytelling.
			Vygotsky	Storytelling as a culturally mediated learning tool.

Process of Learning



Notes

Pedagogical Insight: In foundational and Preparatory Stage classrooms, experiential learning fosters critical thinking, problem-solving, and social skills. It supports different learning styles and helps children build conceptual understanding through interaction, collaboration, and experimentation. This approach aligns well with child-centric and constructivist pedagogy, which values the learner’s active role in constructing knowledge.

7.4.2 Building Concept Formation through Experiential Learning

Concept formation is the cognitive process through which learners categorize objects, ideas, or experiences based on shared attributes, supporting the development of abstract thinking (Bruner, 1960). In the primary years, this process is rooted in hands-on exploration, pattern recognition, and language use, aligning with developmental theories of learning (Piaget, 1973; Vygotsky, 1978). Experiential learning further strengthens concept building by engaging children in real-world interactions that encourage questioning, observing, and reflecting, rather than relying solely on verbal instruction (Kolb, 1984; Dewey, 1938). Through observation, documentation, and comparison of experiences, learners gradually construct meaningful understandings of concepts such as change, growth, and cause-effect relationships (Fosnot, 2013; Woolfolk, 2019). This approach not only fosters curiosity and critical thinking but also cultivates

Process of Learning



Notes

essential skills like attention to detail and reflective thinking (Schunk, 2020). Key components—such as observing changes (Dewey, 1938), documenting experiences (Malaguzzi, 1996), and making sense of patterns over time (Kolb, 1984)—are central to developing deeper conceptual understanding. Well-designed classroom activities grounded in these principles support inquiry-based, joyful, and developmentally appropriate learning (Vygotsky, 1978; Santrock, 2018).

Table 7.4 Classroom Activities for Concept Formation through Experiential Learning

Activity	Concept Focus	Objective	Material	Steps	Learning Theory aligned
Sorting and Classifying with Natural Materials	Classification and grouping	Learners understand how to group items based on similarities and differences.	Leaves, seeds, stones, sticks	Children collect items from the school garden. Ask them to sort them by shape, size, texture, or colour. Discuss why and how they grouped the materials.	Piaget's Constructivism: Children build knowledge by manipulating and sorting objects. Bruner's Enactive Stage: Hands-on engagement supports cognitive development.

Water Explorers – Solids vs. Liquids	States of matter	Understand the physical properties of solids and liquids through observation.	Ice cubes, water, sponge, pebbles	Place items in water and observe behaviour (e.g., floating/sinking, melting). Children describe what happened and why. Record observations in drawings or simple sentences.	Kolb's Experiential Learning: Children observe and reflect, form concepts, and apply understanding in new contexts. Vygotsky's Sociocultural Theory: Collaborative observation and guided discussion support conceptual growth.
Measuring with Non-Standard Units	Measurement	Grasp the idea of length using informal tools.	Hand spans, paper-clips, blocks	Ask children to measure classroom objects using hand spans or blocks. Compare results and discuss which units were easier/more accurate.	Constructivism: Learners actively construct knowledge by experiencing measurement in context. Bruner's Iconic Representation: Visual representation of measurement helps in abstract thinking

Process of Learning



Notes

Process of Learning



Notes

Role Play – Buying and Selling	Money, addition, and subtraction	Use real-life simulation to understand numerical operations and currency	Play money, classroom store setup	Set up a small “market” with labeled items. Children act as shopkeepers and customers. Use addition/subtraction to calculate transactions.	Vygotsky: Social role-play with peer interaction builds higher-order thinking. Kolb: Concrete experience (role play) is followed by discussion and abstraction.
Mapping the Classroom	Spatial awareness and early geography	Learn to read and create simple maps	Chart paper, drawing tools, markers	Walk around the classroom and identify key spaces. Children create a map showing furniture and other objects. Introduce terms like “left,” “right,” “behind,” “beside.”	Bruner’s Iconic to Symbolic Stage: Moving from visual representations to abstract spatial concepts. Constructivism: Active involvement leads to better understanding of spatial relationships.

Teacher’s Role in Concept Formation through Experiential Learning

- Design meaningful contexts for children to explore concepts.
- Facilitate reflection through open-ended questioning (e.g., “*What did you notice?*”, “*Why do you think that happened?*”).
- Encourage peer learning and scaffold children’s ideas.
- Document progress using observation notes and learning journals.

Pedagogical Insight—Encourage children to use multiple modes of expression—words, pictures, or stories—to record their findings. This accommodates different learning styles and abilities and allows children to communicate their understanding in their preferred way. Reinforce that every observation is valuable, even if it is small or simple.

**CHECK YOUR PROGRESS 7.2**

1. Experiential learning primarily involves:
 - a) Memorising facts and definitions
 - b) Learning through lectures and notes
 - c) Learning by doing and reflecting on actions
 - d) Copying examples from textbooks
2. According to Kolb's Experiential Learning Cycle, which of the following is not a stage?
 - a) Concrete experience
 - b) Abstract conceptualisation
 - c) Passive observation
 - d) Reflective observation

Ans. 1-c; 2-c

7.5 PLAYFUL CLASSROOMS – TOY-BASED PEDAGOGY FOR JOYFUL LEARNING

7.5.1 Toy-Based Pedagogy in Primary Classrooms-Why Toy Matters

Toy-based pedagogy is a structured educational approach that uses toys and manipulatives intentionally to achieve specific learning outcomes, distinguishing it from unstructured play-based learning (UNESCO, 2022). Rooted in experiential and constructivist theories, it engages children in active, hands-on exploration that supports concept formation, problem-solving, and contextual understanding (Kolb, 1984; Piaget, 1973; Fosnot, 2013). This method makes abstract ideas more tangible (Bruner, 1960), enhances multi-sensory engagement (Woolfolk, 2019), and encourages creativity, persistence, and collaboration (Vygotsky, 1978; Santrock, 2018). Effective toy-based pedagogy involves careful selection of age-appropriate, culturally relevant toys, integration across subjects, and guided reflection to deepen understanding (Piaget, 1973; NCERT, 2021; Schunk, 2020).

Process of Learning



Notes

Process of Learning



Notes

Table 7.5: Types of Toys Suitable for foundational & Preparatory Classrooms

Type	Purpose	Examples
Manipulatives	Visualizing math and logic	Beads, number rods, abacus, interlocking cubes
Mechanical Toys	Understanding motion, cause-effect	Gears, pulleys, spinning tops
Balancing Toys	Exploring measurement and weight	Traditional balance scales, lever kits
Sequencing Toys	Building logical thinking, language skills	Picture puzzles, story sequence cards
Patterning Toys	Enhancing observation and early math	Rangoli sets, lacing patterns

Toy based concept-building activities offer engaging, hands-on ways for children to grasp abstract ideas through manipulation, exploration, and discovery. These activities integrate traditional and locally available toys with curriculum concepts, encouraging cognitive, social, and motor skill development. Teachers can use them to reinforce topics in math, language, EVS, and more—while promoting curiosity, collaboration, and joyful learning.

Table 7.6: Sample Activities for Toy-Based Concept Building

Activity	Concept	Toy Used	Steps	Learning Outcome
Math with Beads and Number Rods	Counting, addition, place value	Co-lour-coded beads, number rods	Children use beads to count and form number groups (e.g., tens and ones). Create number stories (e.g., 3 beads + 2 beads = 5 beads).	Understand place value and perform basic arithmetic.
Story Sequencing with Picture Tiles	Narrative structure, sequencing, vocabulary	Picture story cards or tiles	Shuffle and distribute story tiles. Children work in groups to arrange them in logical order and narrate.	Develop story-telling, logical flow, and sentence formation.

Traditional Balance Scale for Measurement	Weight comparison, estimation	Home-made balance scale (hanger + cups)	Children compare different classroom objects. Predict and test which object is heavier or lighter.	Grasp non-standard units of measurement.
Spinning Tops for Motion and Observation	Rotation, speed, symmetry	Traditional spinning tops	Spin tops made from different materials (paper, wood, plastic). Observe which spins longer/faster and why.	Explore motion and material properties.
Pattern Creation with Rangoli Peg Sets	Repeating patterns, symmetry	Coloured pegs or rangoli toy sets	Children create patterns using shapes and colours. Extend or mirror a given pattern.	Identify and extend visual patterns.

Process of Learning



Notes

Teacher's Role in Toy-Based Pedagogy :

- Curator: Select and prepare educational toys based on curriculum and learner needs.
- Facilitator: Pose guiding questions and allow independent exploration.
- Observer: Document student strategies and conceptual development.
- Reflector: Encourage learners to articulate their thought processes post-activity.

7.5.2 Toy-Based Pedagogy: Alignment with Learning Theories and Child Development

Toy-based pedagogy is a purposeful, structured approach that uses educational and traditional toys to support concept development, critical thinking, and holistic learning. When grounded in child development principles and learning theories, it becomes an effective tool for fostering cognitive, social, emotional, physical, and creative growth. Toys like puzzles and number rods make abstract concepts tangible, enhancing memory and reasoning (Piaget, 1973; Bruner, 1960), while group play activities build empathy and cooperation (Vygotsky, 1978; Santrock, 2018). Language skills are strengthened through interactive tools such as puppets and flashcards (Woolfolk, 2019), and physical development is supported by manipulative toys like lacing beads and stacking

Process of Learning



Notes

blocks (NCERT, 2021). Creative thinking is encouraged through open-ended materials like clay and building blocks (UNESCO, 2022; Kolb, 1984). Effective implementation involves selecting developmentally appropriate, culturally relevant toys, integrating them across subjects, and encouraging reflection to deepen learning (NCERT, 2021; Vygotsky, 1978).

Table 7.7: Toy-based Pedagogy and Alignment with Key Learning Theories

Learning Theory	Stage of Development	How Toys support	Example
Jean Piaget's Cognitive Development Theory	Preoperational (2–7 years), Concrete Operational (7–11 years)	Toys like building blocks, puzzles, and sequencing games support symbolic thinking and concrete operations. Sorting toys and pattern sets promote classification and seriation, key aspects of logical thinking.	Using counting beads to teach number sense supports the transition from intuitive to logical reasoning.
Lev Vygotsky's Sociocultural Theory	Zone of Proximal Development (ZPD), scaffolding through social interaction	Collaborative play with toys encourages peer learning and guided discovery. Teachers act as facilitators, supporting children just beyond their current level of competence.	Playing a math game with dice and number boards allows students to learn from peers and teacher prompts.
Howard Gardner's Multiple Intelligences Theory	All Stages	Toys cater to different intelligences—musical toys for musical intelligence, construction sets for spatial intelligence, role-play kits for interpersonal intelligence.	Puppet toys enhance linguistic and interpersonal intelligence during storytelling
Jerome Bruner's Constructivist Theory	All stages	Manipulatives and toys promote active learning, where children discover principles through exploration.	Balancing toys help children intuitively grasp the concept of weight and equilibrium.

Maria Montessori's Philosophy	Preoperational (2–7 years), Concrete Operational (7–11 years)	Montessori-aligned toys (bead chains, sandpaper letters) support independent, self-correcting learning.	Use of color-coded rods to build an understanding of length and sequencing independently.
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Toy-based pedagogy, when guided by sound learning theories and child development principles, can significantly enhance classroom engagement and conceptual clarity. It transforms passive learning into joyful, active exploration, supporting children's holistic growth and long-term learning outcomes.

Table 7.8: Example Activities aligned with Learning Theories

Toy	Learning Theory	Developmental Domain	Sample Activity
Number beads	Piaget (Concrete Operational)	Cognitive	Create addition stories with beads
Puppet set	Vygotsky (Social Interaction)	Language, Social	Act out a dialogue from a story
Balancing toy	Bruner (Discovery learning)	Cognitive, Motor	Compare weight of classroom items
Rangoli pegs	Gardner (Spatial Intelligence)	Creative, Math	Create and extend visual patterns
Traditional spinning tops	Montessori (Sensorial Learning)	Motor, Observation	Observe spin time and discuss material differences



CHECK YOUR PROGRESS 7.3

- Toy-based pedagogy is effective because it:
 - Reduces the teacher's workload
 - Makes learning joyful and conceptually grounded
 - Keeps children busy without needing instruction
 - Replaces textbooks entirely
- Traditional Indian toys can support learning by:
 - Encouraging rote memorisation
 - Promoting passive observation

Process of Learning



Notes

Process of Learning



Notes

- c) Encouraging critical thinking and fine motor skills
- d) Being used only for free play

Answer: a-b; 2-c

7.6 FOUNDATIONS OF INTEGRATED LEARNING

Integrated learning is an instructional approach that connects concepts and skills from multiple subjects into unified, meaningful learning experiences, helping children understand content in a holistic and relevant way (Drake & Reid, 2018). Rather than teaching subjects in isolation, it revolves around themes or real-life contexts, fostering deeper engagement—especially in the primary years where contextual and experiential learning is most effective (Fogarty, 1991; NCERT, 2022). Forms of integration include thematic, interdisciplinary, and art- or inquiry-based approaches, which blend cognitive, creative, and social learning—for example, teaching math through stories or exploring seasons with songs, drawings, and science (UNESCO, 2022; Kolb, 1984). This approach supports cognitive development, social-emotional growth, language enrichment, creativity, and physical coordination (Piaget, 1973; Vygotsky, 1978; Woolfolk, 2019; Malaguzzi, 1996; Santrock, 2018). Endorsed by NEP 2020 and NCF-FS, integrated learning promotes joyful, play-based, and culturally rooted education aligned with children’s natural curiosity and developmental needs (MoE, 2020; NCERT, 2022).

Several subjects in the primary curriculum can be meaningfully integrated. Table 7.9 presents an overview with examples and related learning theories:

Table 7.9: Integration of Subjects with each other -Learning Integrated Meaningfully

Subject	Integration with other subjects	Example	Associated Learning theory
The World Around Us (TWAU)	Language, Math, Art	A project on “<i>Plants Around Us</i>” may include: Language: Writing a short story about a tree Math: Measuring plant growth Art: Making leaf rubbings or plant collages	Constructivism (Piaget) – children construct knowledge through hands-on exploration. Inquiry-Based Learning – children ask questions and investigate answers

Mathematics	Music, Movement, Stories	Learning patterns through musical beats or measuring distance through a storytelling adventure	Multiple Intelligences (Gardner) – integrating music and bodily-kinesthetic activities. Bruner’s Spiral Curriculum – building on concepts through varied experiences
Language	EVS, Art, Drama	Reading a folk tale and acting it out while exploring the culture and geography of the setting	Vygotsky’s Social Constructivism – language learning through social interaction. Dramatic Play – supports language development and narrative skills
Art Education	All subjects	Creating puppets to tell a science story, drawing diagrams in math, or illustrating poems	Dewey’s Experiential Learning – learning through doing and reflecting. Reggio Emilia Approach – child-led, art-rich environments for exploration

Process of Learning



Notes

7.6.1 Artful Thinking: Art Forms as Learning Tools

Art forms—such as visual arts, music, dance, drama, and crafts—are powerful pedagogical tools that support diverse learning styles and promote holistic development in foundational & preparatory stage of education (Eisner, 2002; NCERT, 2022). Drawing and clay work enhance creativity and spatial awareness (Lowenfeld & Brittain, 1987), while music supports memory, listening skills, and emotional balance (Gardner, 1983). Dance and drama foster coordination, empathy, and communication, and crafts build fine motor skills and patience (Vygotsky, 1978; Santrock, 2018). Integrating arts into academics makes learning experiential, sensory-rich, and emotionally engaging, deepening conceptual understanding and boosting motivation (Dewey, 1938; Kolb, 1984; Woolfolk, 2019). This approach aligns with key learning theories such as Gardner’s Multiple Intelligences, Vygotsky’s Social Constructivism, and Dewey’s Experiential Learning, all of which emphasize the value of artistic

Process of Learning



Notes

expression in child development (Gardner, 1983; Vygotsky, 1978; Dewey, 1938).

Table 7.10: Art Forms Mapped to Subjects, Learning Outcomes and Learning Theories

Art Form	Linked Subjects	Learning Outcomes Supported	Associated Learning Theories
Visual Arts (drawing, painting, clay, collage)	TWAO, Math, Language	- Observes and represents patterns, plants, seasons - Describes experiences visually - Uses symbols and visuals to communicate ideas	- Piaget's Constructivism (symbolic representation) - Gardner's Visual-Spatial Intelligence - Dewey's Experiential Learning
Music and Rhythm	Language, Math, EVS	- Recites poems with rhythm - Recognises patterns and sequences - Expresses moods through music	- Gardner's Musical Intelligence - Bruner's Pattern Recognition - Vygotsky's Social Mediation through songs
Dance and Movement	Physical Education, TWAO, Language	- Demonstrates body awareness and coordination - Interprets emotions or stories through movement - Engages in rhythmic activities	- Gardner's Bodily-Kinesthetic Intelligence - Dewey's Learning by Doing - Embodied Cognition Theory
Drama and Role-play	Language, EVS, Social Studies	- Enacts stories, events, and characters - Practices empathy and communication - Develops vocabulary and sequence of events	- Vygotsky's Zone of Proximal Development (ZPD) - Social Constructivism - Gardner's Interpersonal Intelligence
Crafts and Hands-on Art	Math, TWAO, Art Education	- Measures, folds, and assembles materials - Identifies shapes and patterns - Demonstrates perseverance and attention to detail	- Piaget's Concrete Operational Stage - Montessori Approach - Gardner's Logical-Mathematical Intelligence

This framework empowers teachers to strategically integrate art forms to enrich classroom teaching, build interdisciplinary connections, and cater to diverse learner needs—making learning meaningful, inclusive, and joyful.

7.6.2 Principles and Practice of Guided Inquiry

Guided inquiry is a structured, child-centered approach that encourages learners to explore meaningful questions with active support from the teacher (Kuhlthau, Maniotes, & Caspari, 2015). Unlike free exploration, it provides a clear framework aligned with developmental needs and curricular goals (Bruner, 1960; Woolfolk, 2019). Teachers act as facilitators—offering resources, prompting questions, and guiding reflection without directing outcomes (Santrock, 2018). The typical inquiry cycle includes asking questions, exploring, creating, discussing, and reflecting. Rooted in learning theories such as Vygotsky’s Social Constructivism, Bruner’s Discovery Learning, and Piaget’s Constructivism, guided inquiry promotes active engagement, peer collaboration, and scaffolded learning (Vygotsky, 1978; Bruner, 1960; Piaget, 1973). When designed around real-world, age-appropriate questions—like “*What do plants need to grow?*” or “*How can we reduce waste in our school?*”—this approach nurtures curiosity, critical thinking, and foundational 21st-century skills (NCERT, 2022).

Process of Learning



Notes

Table 7.11: Guided Inquiry in Practice

Component	Description	Learning Theory Alignment
Nature of Guided Inquiry	Structured yet flexible exploration supported by teacher guidance and open-ended tasks	- Vygotsky’s Social Constructivism (learning via scaffolding) - Bruner’s Discovery Learning
Difference from Free Exploration	Free exploration is unstructured and child-directed; guided inquiry is purposeful and scaffolded with embedded learning goals	- Montessori Philosophy (freedom with structure) - Piaget’s Active Learning
Teacher’s Role	- Questioner: Stimulates thinking through open-ended questions - Facilitator: Provides tools, context, and support - Observer: Documents learning and offers feedback	- Dewey’s Experiential Learning - Reggio Emilia Approach (teacher as researcher)
Inquiry Cycle	Ask → Explore → Create → Discuss → Reflect Each stage fosters critical thinking, collaboration, and self-directed learning	- Kolb’s Experiential Learning Cycle - Constructivism (Piaget, Vygotsky)

Process of Learning



Notes

Component	Description	Learning Theory Alignment
Designing Inquiry Questions	- Use age-appropriate language - Ensure relevance to children's daily life and environment - Encourage multiple possible answers and perspectives	- Multiple Intelligences (Gardner) - Bruner's Readiness and Spiral Curriculum

7.6.3 Supporting Age-Appropriate Problem-Solving

Problem-solving in the primary years should align with children's developmental stages, emphasizing hands-on, real-life contexts that match their concrete operational thinking (Piaget, 1973). Activities like conserving water or maintaining classroom cleanliness foster not only logical reasoning but also civic responsibility and emotional growth (Santrock, 2018). Collaborative tasks, such as group discussions and brainstorming, support Vygotsky's social constructivist view, where learning is enhanced through peer interaction and guided support within the Zone of Proximal Development (Vygotsky, 1978). Teachers play a key role in scaffolding learning and encouraging multiple perspectives. Inclusive strategies—like visual aids, modeling, and flexible expression formats—ensure that all learners can engage meaningfully, reflecting Gardner's theory of Multiple Intelligences (Gardner, 1983; Woolfolk, 2019). When problem-solving is developmentally appropriate and differentiated, it nurtures critical thinking, creativity, empathy, and collaboration—skills vital for lifelong learning and responsible citizenship (Bruner, 1960; NCERT, 2022).

7.6.4 CLASSROOM IMPLEMENTATION STRATEGIES

Implementing art integrated and inquiry-based learning effectively in foundational and preparatory classrooms requires thoughtful planning of lessons, materials, space, and time. Lessons should begin with engaging, age-appropriate questions and incorporate artistic activities like collage, drama, or mapping—supporting Dewey's experiential learning, which values learning through doing and reflection (Dewey, 1938). These activities follow an inquiry cycle—ask, explore, create, discuss, reflect—aligning with Piaget's constructivist and Bruner's discovery learning approaches (Piaget, 1973; Bruner, 1961). A flexible, student-centered environment—with open materials and creative displays—encourages autonomy and mirrors the Reggio Emilia and Montessori philosophies, where the environment acts as a “third teacher” (Edwards, Gandini, & Forman, 1998; Lillard, 2005). In low-resource settings,

using local, low-cost materials like clay or natural objects supports cultural relevance and sustainability, consistent with Vygotsky's idea of cultural tools and NEP 2020's emphasis on experiential learning (Vygotsky, 1978; MoE, 2020). Integrating art and inquiry into subject blocks—such as combining math with pattern art or language with puppetry—also caters to diverse learning styles, as proposed by Gardner's theory of Multiple Intelligences (Gardner, 1983), fostering curiosity, collaboration, and deeper engagement.

**CHECK YOUR PROGRESS 7.4**

1. Art-integrated learning helps in:
 - a) Enhancing children's creativity and self-expression
 - b) Teaching art as a separate subject
 - c) Memorising vocabulary only
 - d) Making classes longer and more difficult
2. Which of the following is an example of art integration?
 - a) Teaching TWAU by reading aloud from a textbook
 - b) Using a dance performance to explain plant growth
 - c) Assigning worksheets only
 - d) Reading a poem without expression

Answer: 1-a; 2-b

7.7 FOSTERING LEARNER AUTONOMY IN PRIMARY CLASS-ROOMS

Fostering learner autonomy in foundational and Preparatory classrooms involves creating an environment where children are empowered to make choices, express themselves freely, and take ownership of their learning. By offering structured options, encouraging self-reflection, and supporting self-advocacy, teachers help young learners develop confidence, independence, and responsibility. This approach not only nurtures critical thinking and intrinsic motivation but also aligns with the NEP 2020 vision of creating active, joyful, and self-directed learners.

Process of Learning*Notes*

Process of Learning



Notes

7.7.1 Understanding Learner Autonomy

Learner autonomy in early education refers to a child's growing ability to make choices, set goals, reflect, and express ideas with confidence. It is not about learning alone, but about building agency through guided opportunities for decision-making and self-expression. During primary years, children begin developing the cognitive and emotional skills needed for autonomy, supported by Piaget's view of logical thinking in the concrete operational stage (Piaget, 1973), Vygotsky's emphasis on scaffolding within the Zone of Proximal Development (Vygotsky, 1978), and Montessori's focus on child-led, choice-based learning. These perspectives highlight that autonomy must be nurtured progressively and with support. The NEP 2020 and NCF-FS promote learner agency through joyful, inquiry-driven, and flexible learning environments that encourage self-paced exploration and creativity (MoE, 2020; NCERT, 2022). In practice, fostering autonomy means offering structured choices, encouraging varied modes of expression, and involving children in setting learning goals, building confidence, responsibility, and a strong sense of ownership over their learning.

7.7.2 Creating a Choice Driven Learning Environment

A choice-driven learning environment empowers children by giving them a say in how, what, and with whom they learn, while maintaining clear learning goals. This structured flexibility respects individual interests and learning styles, fostering engagement and motivation. Allowing students to choose between tasks—like drawing, storytelling, or role play—supports diverse intelligences (Gardner, 1983) and nurtures creativity. Flexible tasks with multiple formats enable all learners to participate at their own level, aligning with Vygotsky's Zone of Proximal Development through guided support (Vygotsky, 1978). Making choices helps children develop decision-making and executive function skills, as emphasized by Piaget and Bruner's constructivist theories (Piaget, 1973; Bruner, 1961). The NEP 2020 and NCF-FS highlight learner agency, joy, and inclusion (MoE, 2020; NCERT, 2022), making choice an essential part of a supportive, engaging, and inclusive classroom that builds confidence, independence, and lifelong learning habits.

7.7.3 Encouraging Self Expression & Supporting Self- Advocacy

Self-expression is central to a child's identity, emotional growth, and creativity. In foundational and Preparatory classrooms, fostering self-expression involves offering varied modes—such as art, storytelling, movement, and discussion—for children to share thoughts and emotions beyond conventional academics. This supports holistic development across cognitive, emotional, and physical domains. Howard Gardner's theory of Multiple Intelligences highlights that children express understanding in diverse ways—verbal, visual, kinesthetic—underscoring the need for inclusive expression formats (Gardner, 1983). Likewise, using tools like puppetry, drawing, or digital media helps children communicate complex ideas and feelings, supporting Vygotsky's view of symbolic tools and collaborative expression as foundations of thought and social learning (Vygotsky, 1978). The NCF-FS advocates for joyful, inclusive classrooms where student voice and uniqueness are valued (NCERT, 2022).

Self-advocacy—the ability to express needs, ask for help, and make informed choices—emerges during primary years and requires guided development. Bandura's theory of self-efficacy affirms that when children believe they can influence outcomes, their motivation and persistence grow (Bandura, 1997). Activities like role play, group discussions, and peer collaboration help build communication and social-emotional skills. Teachers nurture this by modelling respectful dialogue, encouraging reflection, and providing tools like goal-setting checklists and emotion cards. Aligned with NEP 2020's child-centred approach fostering self-advocacy strengthens learners' confidence, empathy, and lifelong ability to engage meaningfully in society.

7.7.4 Classroom Strategies and Examples

To foster learner autonomy in the primary classroom, teachers can integrate structured yet flexible strategies such as choice boards, learning menus, and personal learning projects. These tools allow children to select from a range of activities that align with a shared learning goal, enabling expression through their preferred modes. For example, a literacy choice board might include writing a poem, drawing a story map, or acting out a scene. These practices draw on Gardner's theory of Multiple Intelligences, supporting diverse learning preferences, and Bruner's constructivism, which emphasizes active learning through meaningful choices.

Process of Learning



Notes

Process of Learning



Notes

Table 7.12: Classroom strategies in developing learner's autonomy

Strategy	Description	Aligned Learning Theories	Benefits for Learners
Choice Boards & Learning Menus	Students select tasks from a structured menu or grid based on interests and readiness	- Multiple Intelligences (Gardner)- Constructivism (Bruner)- Self-Determination Theory (Deci & Ryan)	Encourages voice and choice, supports diverse intelligences, fosters motivation
Personal Learning Projects	Learners choose a topic to explore and present using formats of their choice	- Constructivism (Piaget, Bruner)- Inquiry-based Learning- Experiential Learning (Kolb)	Builds ownership, curiosity, and creativity; supports deep engagement
Self-Assessment Checklists	Learners evaluate their effort and behaviour with simple, visual tools	- Metacognitive Theory (Flavell)- Self-Efficacy (Bandura)	Promotes reflection, builds responsibility and confidence
Reflection Journals	Students document thoughts, experiences, and learning moments through words or drawings	- Constructivism- Emotional Intelligence (Goleman)- Self-Regulation Theory	Enhances emotional expression, memory retention, and self-awareness
Group Work with Defined Roles	Teams work on tasks with assigned rotating roles (leader, recorder, etc.)	- Social Constructivism (Vygotsky)- Cooperative Learning- Experiential Learning	Develops communication, responsibility, teamwork, and negotiation skills
Shared Decision-Making in Groups	Students make choices collectively on task execution and presentation	- Social Learning Theory (Bandura)- Constructivism- Democratic Education (Dewey)	Builds social agency, negotiation, and inclusive participation

Together, these practices create a classroom environment where autonomy, agency, and collaboration are nurtured through well-designed pedagogical strategies aligned with developmental needs and foundational learning theories.

7.7.5 Challenges and Solutions

Promoting learner autonomy in foundational and preparatory classrooms involves challenges, but these can be addressed with thoughtful strategies. A common issue is balancing freedom with structure—too much choice may confuse young learners. Structured autonomy, such as using choice boards within clear goals, helps maintain focus while offering agency (Vygotsky, 1978; Bruner, 1966).

Ensuring equal participation is another challenge, especially in diverse groups where some children dominate. Teachers can support equity by designing differentiated tasks, rotating group roles, and applying Universal Design for Learning (UDL) to accommodate varied strengths (Gardner, 1983). Tools like self-assessment checklists and goal-setting activities foster reflection and confidence (Bandura, 1997).

Time constraints can also make autonomy feel hard to implement. Yet, small practices—like reflection journals or visual task charts—can be embedded in daily routines. The gradual release of responsibility model (“I do, we do, you do”) supports the development of independence through guided practice (Kolb, 1984).

By applying these strategies, aligned with NEP 2020 and the NCF for the Foundational Stage (NCERT, 2022), teachers can cultivate self-directed learners in inclusive, supportive environments.



CHECK YOUR PROGRESS 7.5

1. Inquiry-based learning encourages children to:
 - a) Follow teacher instructions only
 - b) Memorise fixed answers
 - c) Ask questions and explore possible answers
 - d) Copy solutions from the board
2. A good example of guided inquiry is:
 - a) Dictating notes for children to copy
 - b) Asking students to observe how water changes form when heated
 - c) Explaining everything without letting children ask questions
 - d) Giving only multiple-choice test

Answer: 1-b; 2-b

Process of Learning



Notes

Process of Learning



Notes

7.8 FACILITATING CHILD CENTRED AND FLEXIBLE PEDAGOGY - TEACHER AS FACILITATOR AND CO-EXPLORER

Facilitating child-centered and flexible pedagogy requires the teacher to shift from being a sole knowledge provider to becoming a facilitator and co-explorer in the learning journey. In this role, the teacher creates a supportive environment where children's interests, questions, and ideas guide the learning process. By listening actively, asking open-ended questions, and adapting to learners' needs, the teacher encourages curiosity, critical thinking, and collaboration. This approach nurtures autonomy and helps children construct meaningful knowledge through exploration and reflection, making learning both engaging and relevant.

7.8.1 Principles of Child-Centric Pedagogy

Child-centric pedagogy places the child at the heart of the learning process, emphasising respect for their interests, abilities, and developmental needs. This approach fosters an environment where learners actively construct knowledge, engage emotionally and cognitively, and develop autonomy. Several core principles guide this pedagogy, rooted in foundational learning theories.

1. **Active Learning and Constructivism** - Children learn best when they actively engage with concepts rather than passively receiving information. According to Piaget's Constructivist theory, learners build new understanding based on their prior knowledge and experiences through exploration and hands-on activities. The teacher's role shifts from knowledge transmitter to facilitator who provides meaningful contexts and tools for discovery.
2. **Social Interaction and Vygotsky's Sociocultural Theory** - Learning is inherently social. Vygotsky emphasized the role of social interaction and cultural context in cognitive development, introducing the concept of the Zone of Proximal Development (ZPD)—the difference between what a learner can do independently and with guidance. Child-centric pedagogy incorporates collaborative learning, peer discussion, and scaffolded support to promote higher-order thinking and skill development.
3. **Differentiation and Multiple Intelligences** - Recognizing that each child has unique strengths, interests, and ways of learning is crucial. Howard

Gardner's Multiple Intelligences Theory encourages educators to design activities that cater to varied intelligences—linguistic, logical-mathematical, spatial, bodily-kinesthetic, musical, interpersonal, intrapersonal, and naturalistic. This diversity in approaches ensures inclusivity and nurtures each child's potential.

4. **Motivation and Self-Determination** - Deci and Ryan's Self-Determination Theory highlights autonomy, competence, and relatedness as key drivers of intrinsic motivation. Child-centric pedagogy supports learner autonomy by providing choices, fostering a sense of competence through achievable challenges, and nurturing meaningful relationships within the classroom.
5. **Emotional Engagement and Experiential Learning** - Emotions deeply influence learning. Engaging children emotionally through stories, arts, and real-world connections enhances retention and enthusiasm. Kolb's Experiential Learning Cycle (concrete experience, reflective observation, abstract conceptualization, and active experimentation) underpins child-centric methods that connect feelings with cognition, encouraging learners to reflect and apply knowledge in varied contexts.
6. **Flexibility and Responsiveness** - Child-centric pedagogy values flexibility—adapting plans based on learner responses and interests. Teachers continuously observe, listen, and adjust instruction, embodying the role of a co-explorer alongside children. This responsiveness respects the dynamic nature of learning and supports personalized growth trajectories.

By integrating these principles, child-centric pedagogy creates an empowering, inclusive, and joyful learning environment that aligns with the National Education Policy (NEP) 2020's vision of holistic, learner-focused education.

7.8.2 Strategies for Flexible and Responsive Teaching

Strategies for flexible and responsive teaching involve adapting instruction to meet the diverse needs, interests, and abilities of learners in real time. This includes differentiating tasks, offering choices, using ongoing assessments to guide teaching, and encouraging reflection. By creating dynamic learning groups and asking thoughtful questions, teachers foster deeper understanding and engagement. Such strategies, grounded in constructivism and sociocultural learning theories, empower students to take ownership of their learning while ensuring that teaching remains relevant and inclusive.

Process of Learning



Notes

Process of Learning



Notes

Table 7.13 Strategies for Flexible and Responsive Teaching

Strategy	Description	Aligned Learning Theories	Benefits for Learners
Differentiated Instruction	Tailoring teaching methods, materials, and pace to meet diverse learners' needs and abilities	Gardner's Multiple Intelligences Vygotsky's ZPD	Addresses individual strengths; promotes inclusion and engagement
Formative Assessment & Feedback	Ongoing assessment to inform instruction and provide timely, constructive feedback	Black & Wiliam's Formative Assessment Theory Bandura's Social Learning	Helps learners adjust strategies; promotes self-regulation and motivation
Flexible Grouping	Creating dynamic groups based on interests, skills, or learning goals that change as needed	Vygotsky's Social Constructivism Cooperative Learning	Encourages peer learning; adapts to learners' evolving needs
Choice-Based Activities	Offering students options in tasks, topics, or presentation modes	Deci & Ryan's Self-Determination Theory Bruner's Constructivism	Enhances motivation, autonomy, and ownership of learning
Responsive Questioning	Using open-ended, probing questions tailored to learner responses	Socratic Method Vygotsky's Scaffolding	Stimulates critical thinking and deeper understanding
Reflective Practice	Encouraging learners to think about their learning processes and outcomes	Kolb's Experiential Learning Cycle Flavell's Metacognition	Develops self-awareness and higher-order thinking skills
Adaptation of Learning Environment	Modifying physical or virtual spaces to support diverse learning styles and activities	Universal Design for Learning (UDL) Bronfenbrenner's Ecological Systems Theory	Supports accessibility and comfort; promotes active engagement

**CHECK YOUR PROGRESS 7.6**

1. A teacher who acts as a facilitator:
 - a) Delivers all knowledge directly
 - b) Observes and guides students' learning journey

Process of Learning

- c) Controls all classroom conversations
 - d) Punishes students for mistakes
2. What does “co-explorer” mean in a learning environment?
- a) Teacher and students learn together
 - b) Teacher gives all the answers
 - c) Students explore while the teacher supervises silently
 - d) Teachers explore only when they have free time

Answer: 1-b; 2-a

**7.9 LET US SUM UP**

- Learning is most effective when children actively engage in experiences and reflect on them.
- Real-life tasks, field explorations, role plays, and classroom experiments support concept-building.
- Encourage critical thinking, collaboration, and application of knowledge.
- Toys—especially traditional and local ones—serve as joyful, low-cost learning tools. They promote creativity, fine motor skills, numeracy, spatial awareness, and social learning.
- Emphasise the role of play in cognitive and emotional development.
- Visual, performing, and literary arts can deepen understanding, expression, and imagination.
- Encourage learning through drawing, drama, music, dance, and storytelling.
- Promote cultural relevance, emotional connection, and engagement.
- Children are natural questioners and problem solvers.
- Inquiry-based learning fosters observation, exploration, experimentation, and reasoning.
- Teachers guide students in asking questions and designing age-

*Notes*

Process of Learning



Notes

appropriate investigations.

- Children develop confidence and ownership when they make choices and express ideas.
- Classrooms should allow voice, choice, and independence in learning paths.
- Foster intrinsic motivation, self-regulation, and lifelong learning habits.
- The teacher's role shifts from instructor to guide, listener, and co-learner.
- Create flexible, child-centric learning environments.
- Encourage shared discovery, mutual respect, and responsive teaching.
- Concept-building is best supported through active, sensory-rich, and contextual learning.
- Hands-on activities should be simple, inclusive, and linked to real-life themes.
- Integrate art, play, and inquiry into lessons across subjects.
- Focus on what children know, understand, and can do after learning.
- Include cognitive, social-emotional, and creative dimensions.
- Align with NIPUN Bharat and Foundational Literacy and Numeracy (FLN) goals.

**7.10 END EXERCISES**

1. How does experiential learning support cognitive and emotional development in young learners?
2. What challenges might a teacher face when implementing experiential learning? Suggest solutions.
3. Design a simple classroom activity using a traditional toy that encourages both play and concept learning.
4. How does toy-based pedagogy promote joyful learning and engagement?

Process of Learning

5. Identify challenges children may face during guided inquiry and ways a teacher can support them.
6. How does promoting learner autonomy contribute to a child's overall development?
7. Discuss how child-centric and flexible pedagogy can be integrated into daily classroom routines.
8. Why is experiential learning effective for concept formation in primary classrooms?
9. How can teachers balance providing freedom of choice with maintaining classroom structure for young learners?



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Process of Learning



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